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CPSM Project: **Pose Imitation of Human Motion by a Simulated Humanoid Robot**

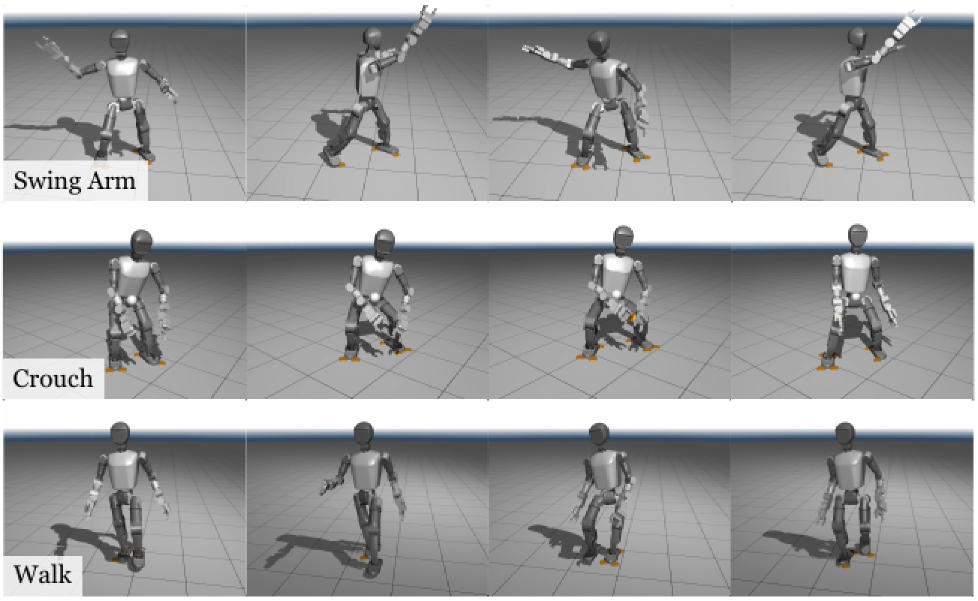

Fig. 1: (from [1])

Background of the project:

Humanoid robots are a field of scientific interest as humanoid robots promise to perform in environments designed for humans without costly modifications. Bipedal locomotion and action however requires intelligent steering to maintain balance. Recent approaches see the usage of machine learning to meet this challenge by mimicking human movements [1, 2].

Project short description:

Motions and poses of a single human are being tracked and replicated by a simulated humanoid robot. The motions and poses are being tracked using only visual information. For this purpose a tripod mounted high quality camera providing a video stream of the human is to be used. A controller will process the video data to steer the motors of the humanoid robot to mimic the human in the Webots simulation environment.

References:

[1] Ze et. al., TWIST: Teleoperated Whole-Body Imitation System, CoRL 2025, May 2025, https://doi.org/10.48550/arXiv.2505.02833

[2] Tao et. al., Visual Perception Method Based on Human Pose Estimation for Humanoid Robot Imitating Human Motions, CCRIS 21 Pages 54 - 61, October 2021, https://doi.org/10.1145/3483845.34838